



CPSC 100

Computational Thinking

Human Computer Interaction

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Department of Computer Science
University of British Columbia

Agenda

- Frustrating Activity
- HCI Usability Components
- HCI Usability Heuristics



Learning Goals

After this **today's lecture**, you should be able to:

- **Understand** and describe the **first five usability heuristics** proposed by Jakob Nielsen.
- **Explain why** each heuristic is important in designing usable systems.
- **Illustrate** how each heuristic appears (or is violated) in real-world interfaces using provided examples.
- **Compare and contrast** different heuristics using concrete UI examples to determine which are adhered to or violated.

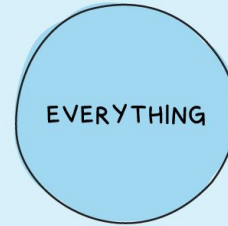
Course Admin



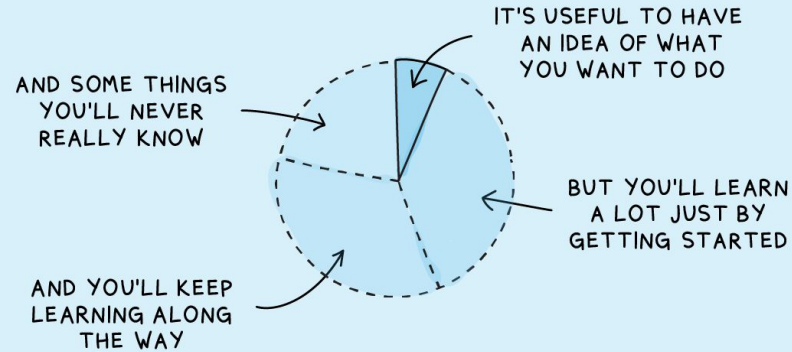
Course Admin

- Project Milestone 1 submission is open on PrairieLearn !
- Due Date is Tuesday October 14th at 6 PM
 - (48-hr grace period applies)

WHAT I THOUGHT I NEEDED TO KNOW TO GET STARTED



THE TRUTH



LIZ FOSSLIE



HCI: Usability Components

1. Learnability
2. Efficiency
3. Memorability
4. Errors
5. Satisfaction



HCI: Usability Components

- **Learnability:**
 - How easy is it to learn task the first time?
- **Efficiency:**
 - How quickly can tasks be done (post-learning)?
- **Memorability:**
 - How easy is it to re-establish proficiency after being away?
- **Errors:**
 - How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
- **Satisfaction:**
 - How pleasant is it to use the design?



Activity

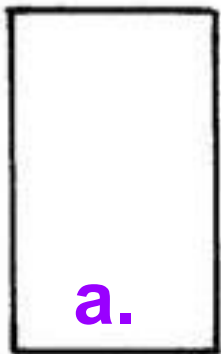


Intro to HCI Activity

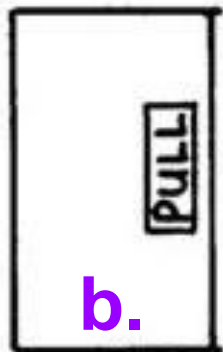
- **Answer on PrairieLearn Class Activity for today**

1. Think of a technological interaction from last week that irritated you.
2. Draw/visualize it (to the best of your ability)
3. Explain exactly HOW it failed for you. Depict activity, tasks, interactions.

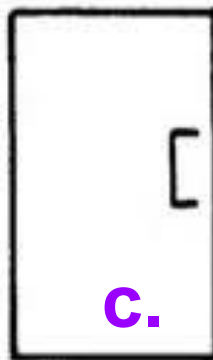
PLAIN DOOR



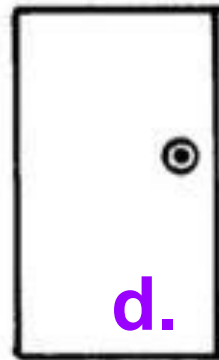
LABELED DOOR



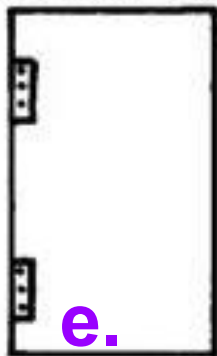
HANDLE DOOR



KNOB DOOR



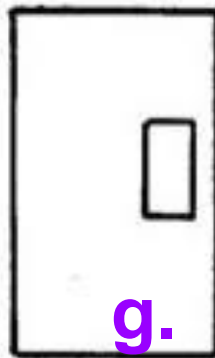
HINGE DOOR



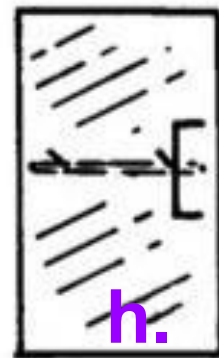
BAR DOOR



PANEL DOOR



GLASS DOOR



Q: How does this door work?

- A. Push to the left
- B. Push to the right
- C. Pull on the left
- D. Pull on the right
- E. Slide it along

PLAIN DOOR



- ☐ Push ☐ Left side
- ☐ Pull ☐ right side
- ☐ slide it along

Q: How does this door work?

- A. Push to the left**
- B. Push to the right**
- C. Pull on the left
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Q: How does this door work?

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- ☐ Pull ☐ right side
- ☐ Slide it along

Q: How does this door work?

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Learning Goals

What are Usability Heuristics?

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What are Usability Heuristics?

What is Usability?

HCI: Usability

- Quality attribute
 - Assesses how easy user interfaces are to use
 - Improving ease-of-use during the design process
- Defined by 5 quality components

HCI: Usability Components

Recall

1. Learnability
2. Efficiency
3. Memorability
4. Errors
5. Satisfaction

What are Heuristics?



What is a Heuristic?

Practical method or guideline

- Used to facilitate
 - Problem-solving
 - Learning
 - Discovery
- Often referred to as a **"rule of thumb"**

Usability Heuristics

10 Usability Heuristics (Nielsen, 1993)



Visibility of
System Status

1



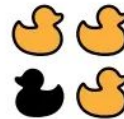
Match Between
System & Real World

2



User Control
And Freedom

3



Consistency
And Standards

4



Error
Prevention

5



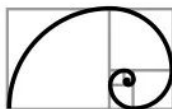
Recognition
Rather Than Recall

6



Flexibility And
Efficiency of Use

7



Aesthetic And
Minimalistic Design

8



Help Users
With Errors

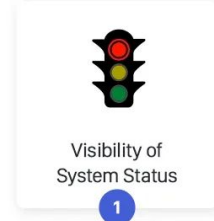
9



Help And
Documentation

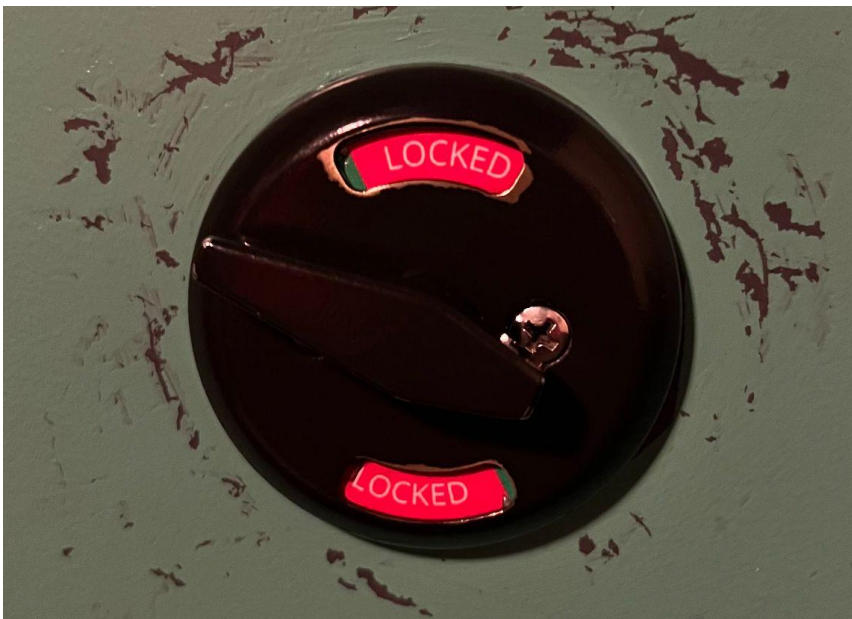
10

1. Visibility of system status



- Keep users informed about what is going on
 - e.g., what page they are on and what part of a process
 - (# of steps left)
 - e.g., you are working in offline mode (connection lost) provide appropriate feedback
- About what system is doing, and how input is being interpreted

Example: Washroom door lock status



Feedback



Visibility of
System Status

1

- Does the user know what happened?
- Sending information back to the user about what has been done
- Includes sound, highlighting, animation and combinations of these
 - e.g. when screen button clicked on provides sound or red highlight feedback:

Previous → “ccclchhk”

Previous → Previous



Step 6/10



Set Your Password

In order to keep your account safe you need to create a strong password.

PASSWORD

CONFIRM PASSWORD

YOUR PASSWORD MUST CONTAIN

- ☐ Between 8 and 20 characters
- ☐ 1 upper case letter
- ☐ 1 or more numbers
- ☐ 1 or more special character

Next Step



Step 6/10



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Next Step



Visibility of
System Status

1



2. Match System + Real world



Match Between
System & Real World

2

- Terminology in user's language language from user's perspective
 - “you have bought...” VS “we have sold you...”
 - Use common words, not “techno-jargon”
- Error messages and feedback refer to user objects
- Avoid saying “you’ve entered an illegal input”



Match Between
System & Real World

2





3. User control + Freedom



User Control
And Freedom

3

- Easy to abort: cancel buttons
 - e.g., being able to cancel/undo order
- Easy to undo
 - e.g., being able to go back to previous step (s)
- Easy to make changes
 - e.g., removing items from a shopping cart
- Users (even *experts*) will make errors!

Example: User control + Freedom



User Control
And Freedom

3

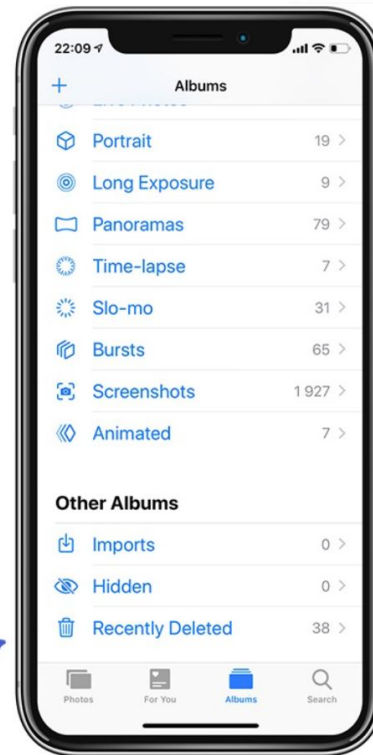


Undo the unwanted action.



Exit the navigation anytime.

Recover your accidentally deleted files.





4. Consistency + Standards



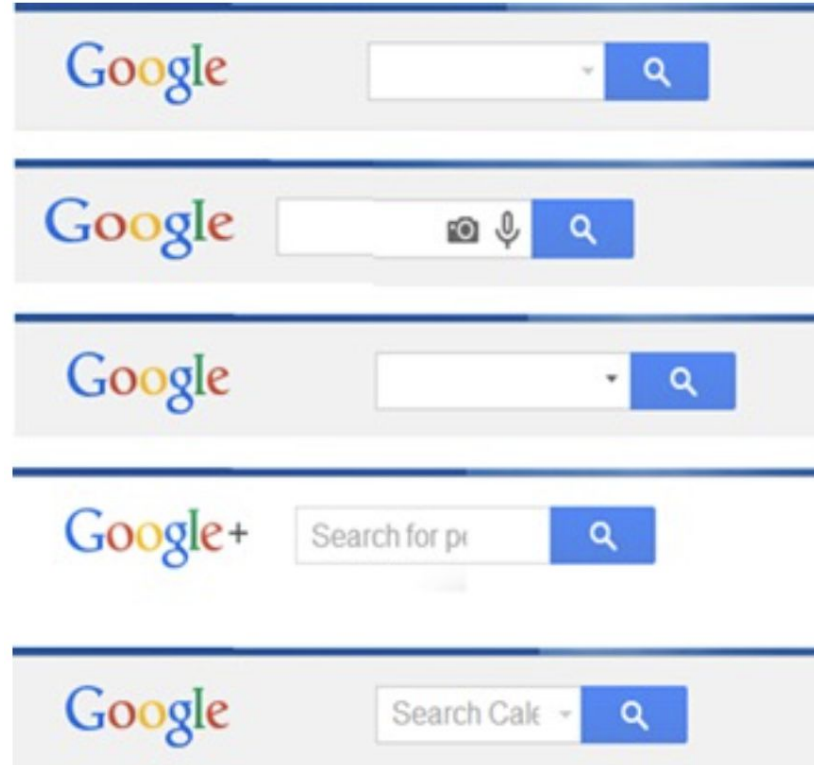
- Same commands always have the same effect
 - (e.g., ctrl+c)
- Locations for information, names of commands give the user a mental model of the system
- Size, location, colour, wording, function, sequencing
- Following standards helps
 - Web: use templates or css, style guides seems easy, but often not followed

Example: Google's Search bar

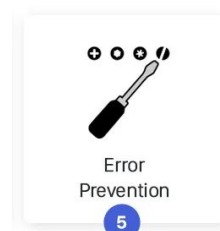


Consistency
And Standards

4



5. Error Prevention



- Constraints
 - Remove or gray-out illegal choices
- Auto-fill information from before
- Confirmation
 - Before making irreversible changes
 - Provide a way to "undo"
- Do not use colours to illustrate danger
 - Colour blindness / accessibility

Example: Alerts/Confirmation



Error
Prevention

5

It seems like you have forgotten to attach a file.

You wrote "are attached" in your message, but there are no files attached. Send anyway?

Cancel

OK

10 Usability Heuristics (Nielsen, 1993)



Visibility of
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1



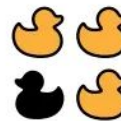
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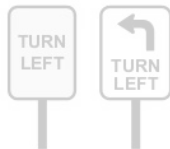
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Help Users
With Errors

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Help And
Documentation

10





Q: Which heuristic does this interface violate?

- A. Visibility of System Status
- B. Match between System & Real World
- C. User Control and Freedom
- D. Consistency and Standards
- E. Error Prevention

Would you mind answering a few questions to help us learn more about you and improve our service?

Continue



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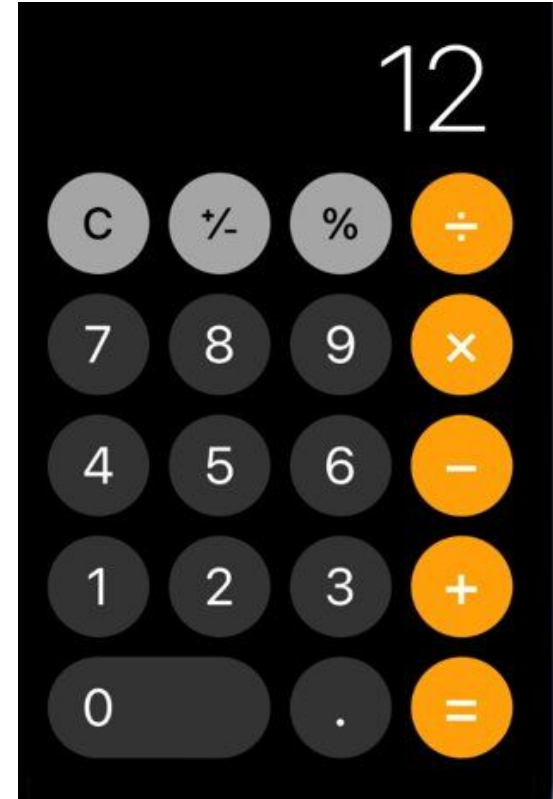




iClicker

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iClicker

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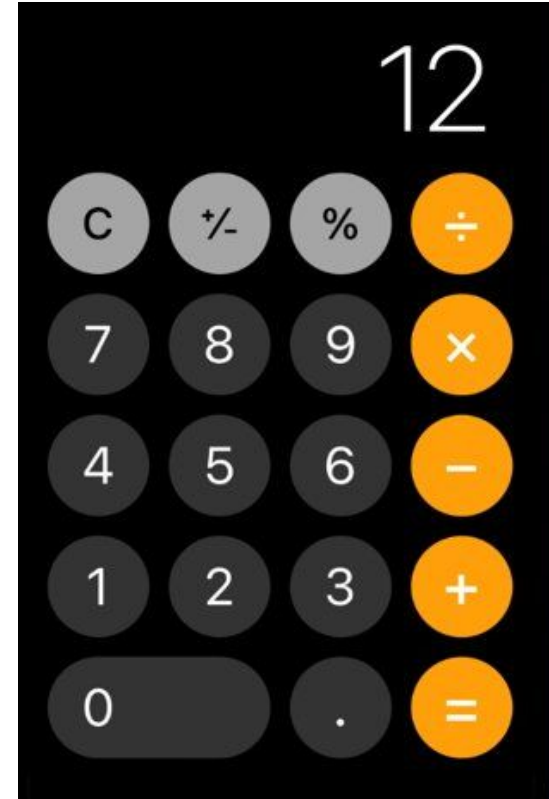
A. Visibility of System Status

B. Match between System & Real World

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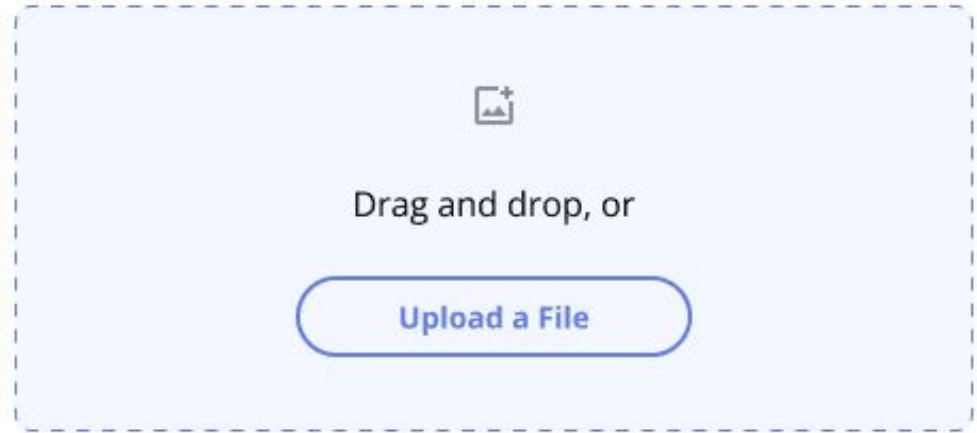
E. Error Prevention





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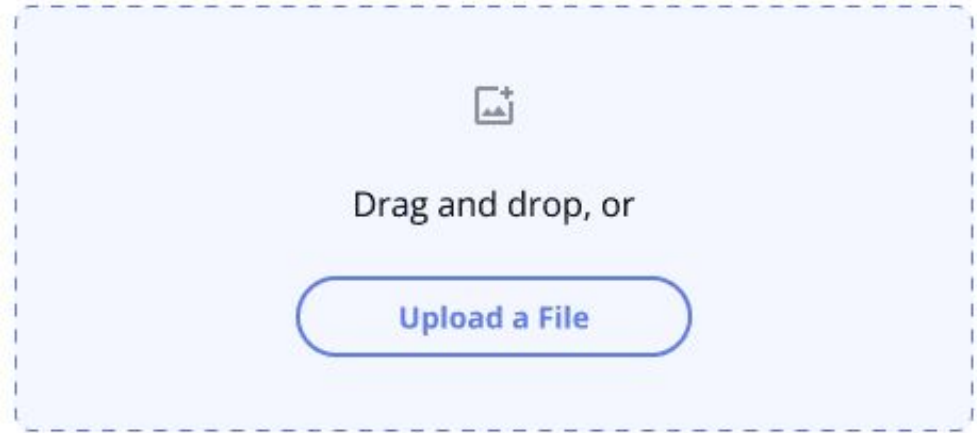


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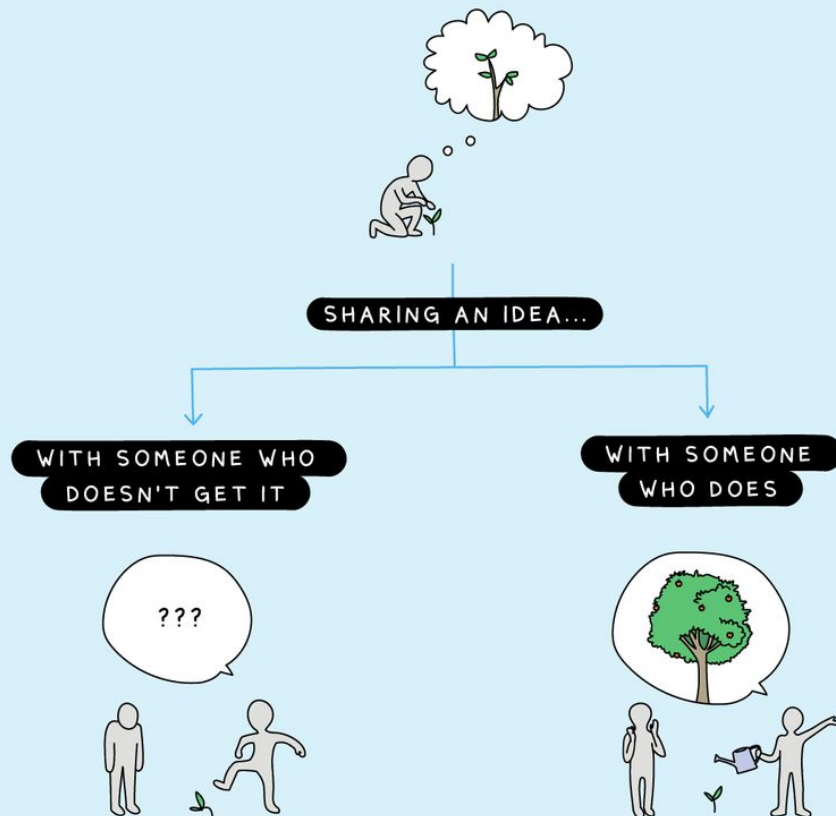


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⚠ The file profile_pic.jpg is not the right size. Upload an image that is 300x300 only.



Wrap up

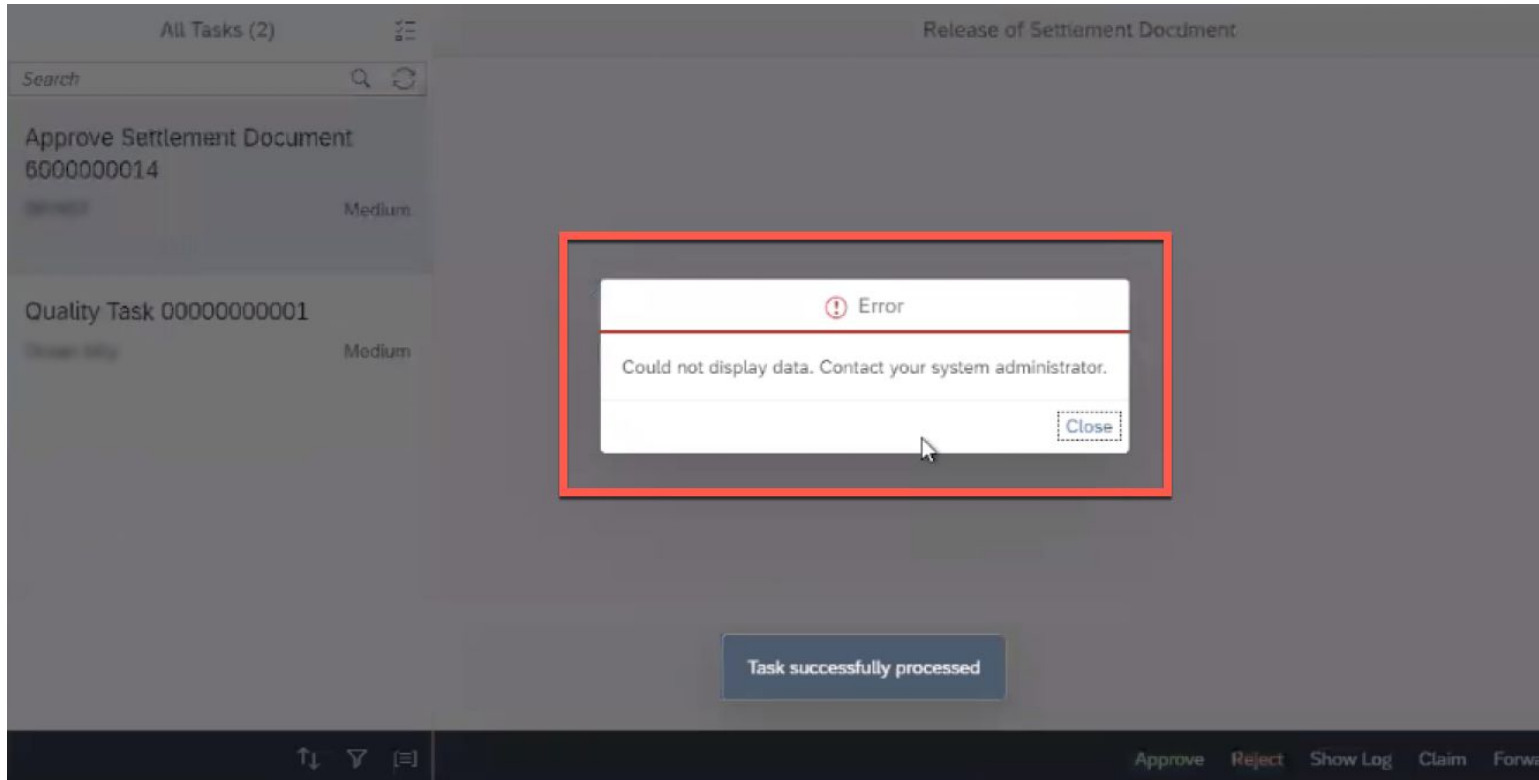
Extra Practice Questions

What is wrong?

Identify the usability heuristic(s) that are **violated** in the following interface:

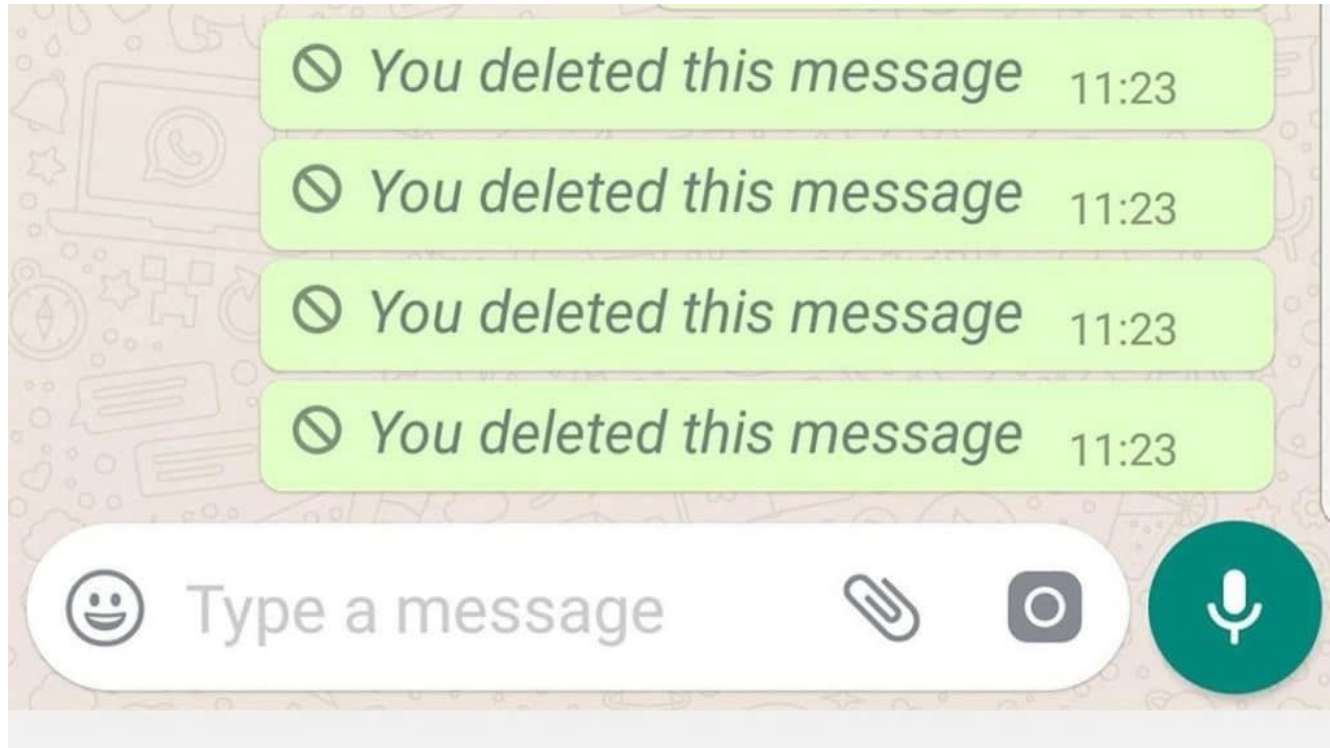


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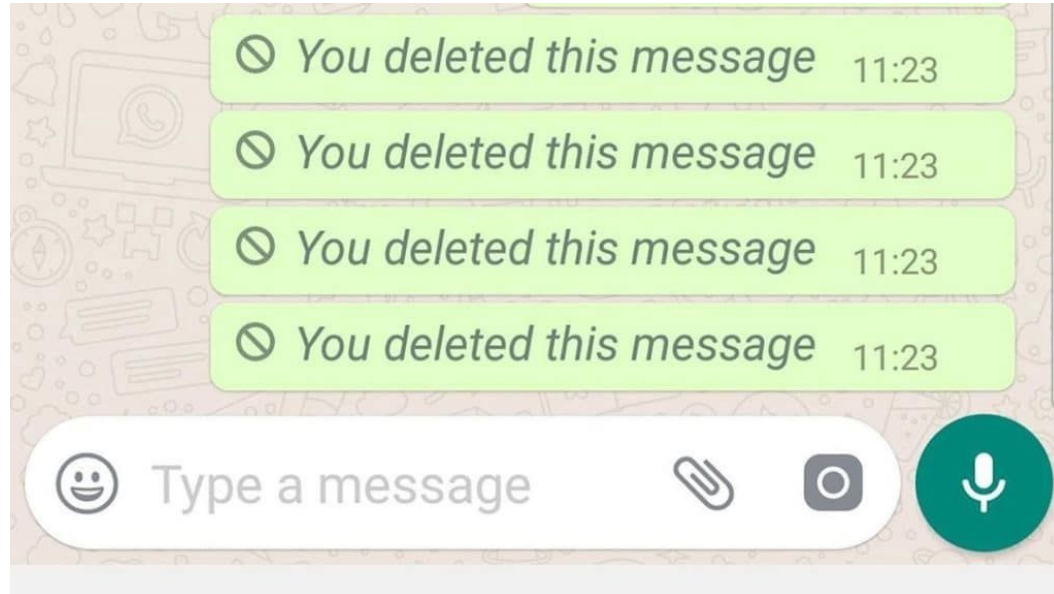
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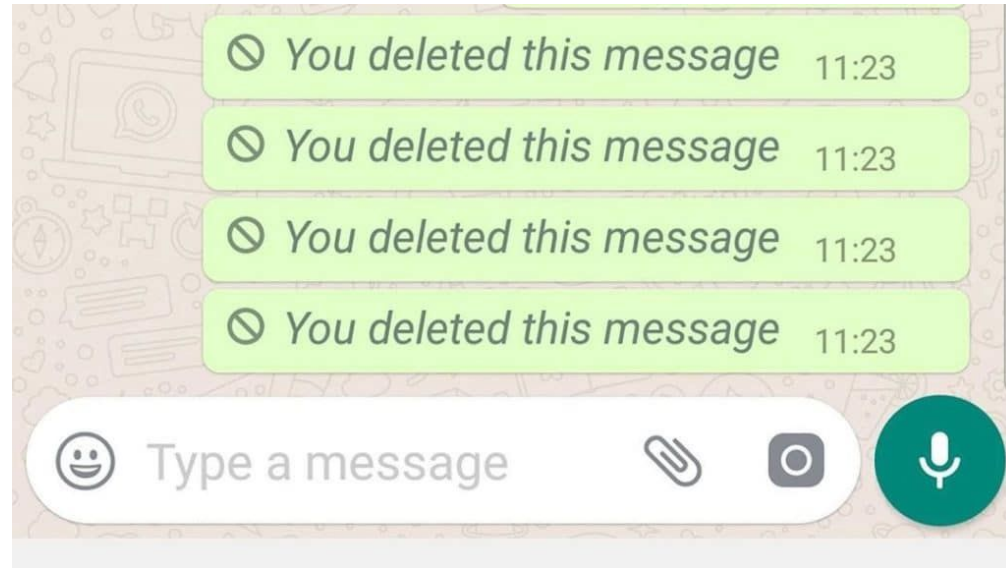




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What is
good?



Identify the usability heuristic(s) that are **satisfied** in the following interface:



Get the Free Guide

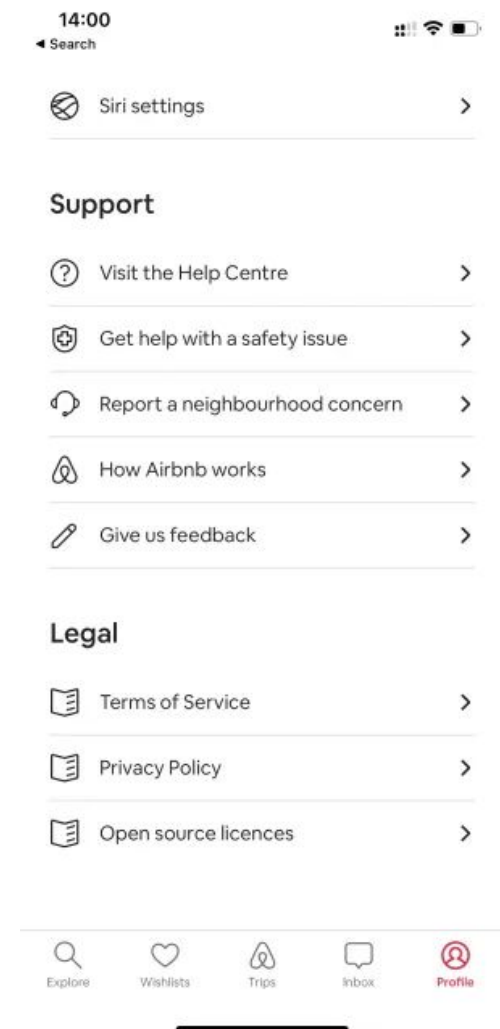
All fields are required.

First Name	Last Name
Clint ✓	Fontanella ✓

Next Step



Identify the usability heuristic(s) that are **satisfied** in the following interface:



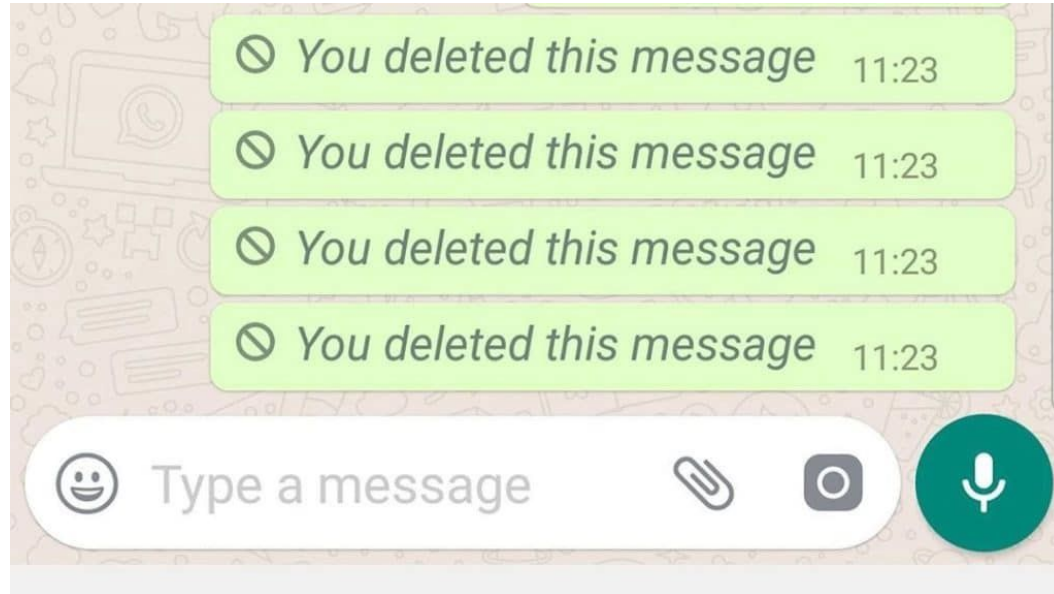
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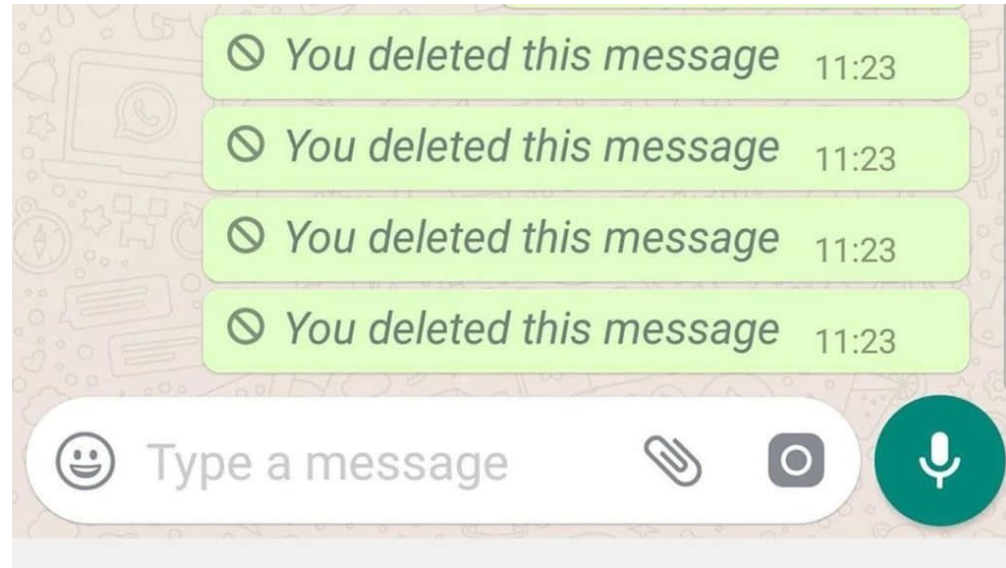




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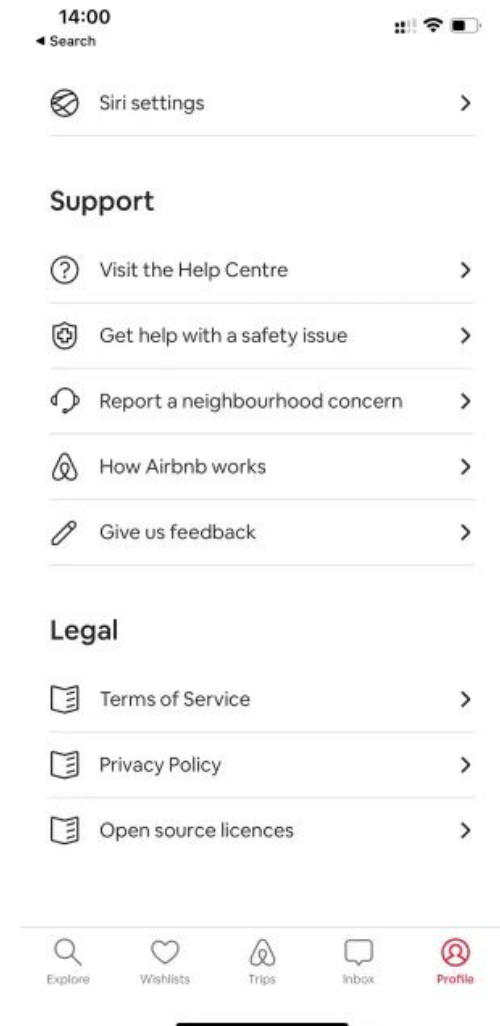
All fields are required.

First Name	Last Name
Clint ✓	Fontanella ✓

Next Step



Identify the usability heuristic(s) that are **satisfied** in the following interface:



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Q: Which heuristic does this interface adhere to?

- A. Recognition Rather Than Recall
- B. Match between System & Real World
- C. User Control and Freedom
- D. Flexibility and Efficiency of Use
- E. Error Prevention



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